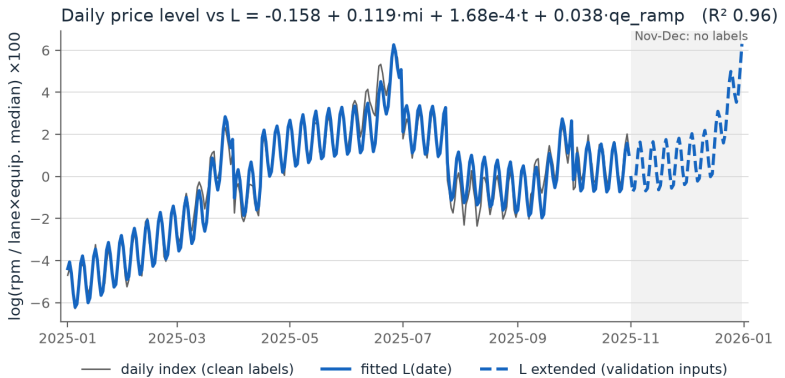
Freight rate prediction

Spotter · Machine Learning Engineer assessment

Denis Bakin

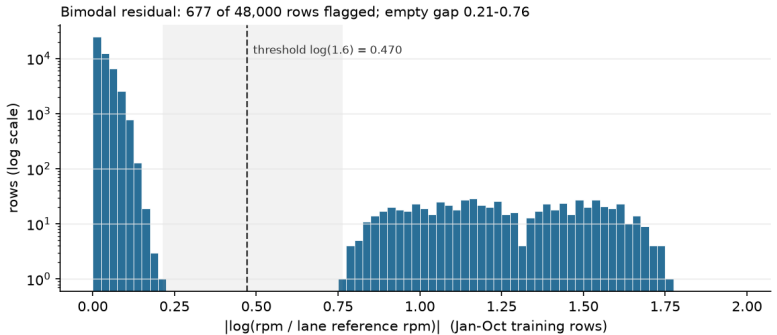
8 October 2026

The price level is market index + trend + quarter-end ramp



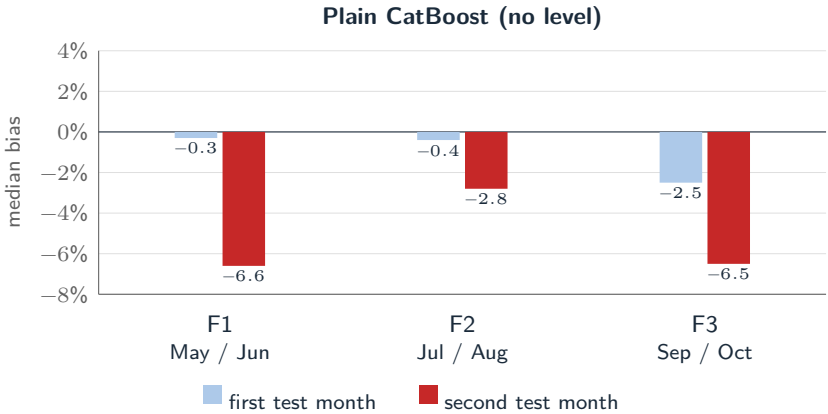
Task: price 12,000 Nov–Dec loads from 48,000 labelled Jan–Oct loads. The metric is undisclosed; 8 validation cities never appear in training.

1.4% of labels are corrupted, and cleanly separable



Corrupted rate ($\times$2.5–5.5 or $\times$0.2–0.45)	677 (1.4%)	dropped from training
Weight: sign flip / cap, floor / missing	292 / 1,204, 32 / 300	abs. value / keep / NaN + flag
Missing `market_index`	374 (0.8%)	that day's mean + flag

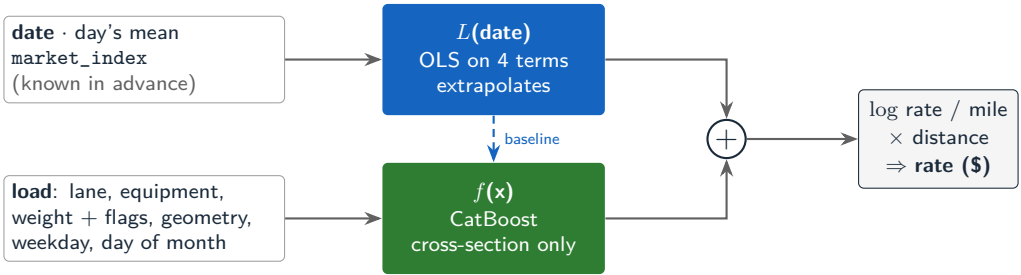
Plain trees nail month 1 and miss month 2 by up to 6.6%



Trees cannot extrapolate a rising price level; adding time t still leaves June at -7.2%.

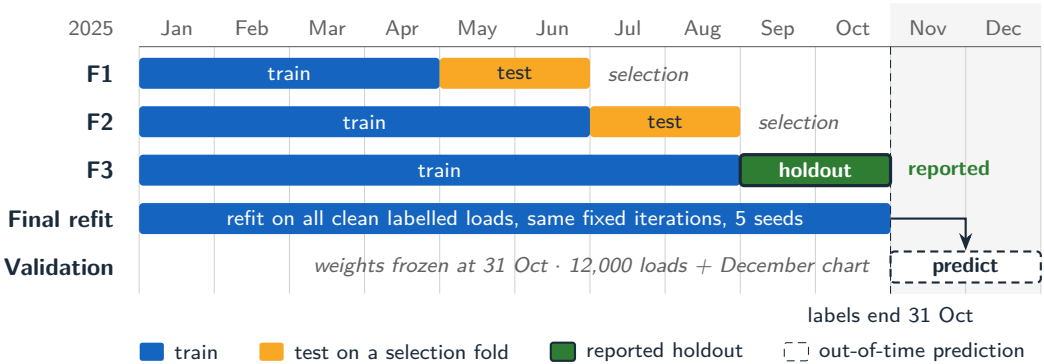
The fix: log(rate / distance) = L(date) + CatBoost(x)

$$\log \frac{\text{rate}}{\text{distance}} = \underbrace{\beta_0 + \beta_{\text{mi}}\,\bar{m}(d) + \beta_t\,t + \beta_{\text{qe}}\,q(d)}_{L(\text{date})\text{: market level, OLS}} + \underbrace{f(\mathbf{x})}_{\text{CatBoost}}$$



L = intercept + day's mean market index + linear trend + quarter-end ramp: all known in advance. CatBoost takes L as its baseline and learns only the lane, equipment, distance and weight structure.

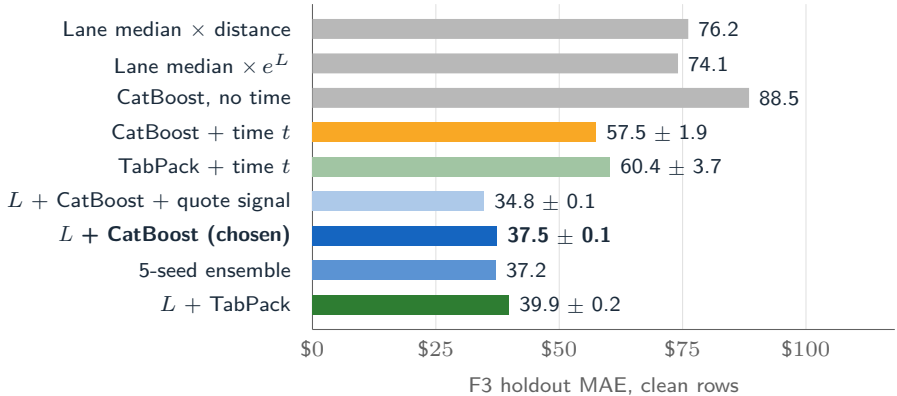
Validate forward in time: select on F1 + F2, report F3 once



Final refit: all 47,323 clean Jan–Oct loads, 5 seeds.

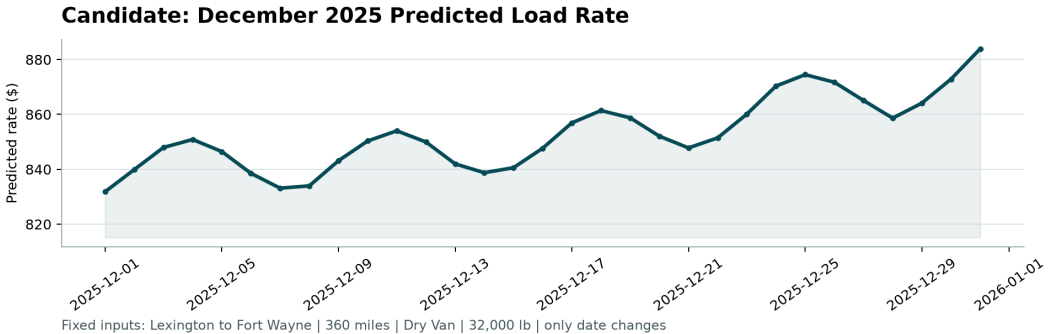
Cold-start check on F3: held-out cities $40.3 MAE vs $34.4 for the rest.

F3 holdout: $37.5 MAE (1.6%), from $76 for the yardstick



Quote-signal variant: better on F3, lost on F1 + F2 ⇒ dropped. Optuna: defaults kept (gain < $0.5).
TabPack: better cross-section, worse level ⇒ blend weight 1.0 on CatBoost.

December: weekly wave + trend + ramp, $832–884



Option A inputs: each day's mean `market_index` from `validation.csv`, not labels.
Weights frozen at 31 Oct. $831.84 on Mon 1 Dec to $883.92 on Wed 31 Dec.

Takeaways, and where they live in the code

- **Validate the way the model is used:** forward in time, F3 scored once.
- **Structure beat tuning:** the level decomposition took F3 MAPE from 2.35% to 1.61%.
- **The learner is not the bottleneck:** TabPack was worth $\approx$ \$1, the decomposition $\approx$ \$20.

File	Key names	Role
`src/freight/outliers.py`	`RpmReference`, `flag_corrupted`	corrupted labels
`src/freight/split.py`	`Fold`, `cold_start_holdout`	forward-time folds, cold start
`src/freight/level.py`	`MarketLevel`, `qe_ramp`	the market level $L(\text{date})$
`src/freight/models.py`	`LevelCatBoostRate`	CatBoost with L as baseline
`src/freight/final.py`	`fit_final`, `predict_december`	final fit, deliverables
`Makefile`	`make all`	tests, train, predict, score